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All statements should be written in C++.

You may find a partial ASCII table at the end of this document.

1. [4 pts] Write the preprocessor directive that you would need to use for the followings.

I. cout << "What is your initials?"; # include < iostream >

II. string name # include < string >

"Headers"

III. abs(x) # include < cmath >

IV. setprecision(2) # include < iomanip >

2. [4 pts] Distinguish each of the variable name if it is legal or illegal. If illegal, explain why.

Variable name		
	Legal?(Y/N)	If illegal, why?
I. tax amount	N	whitespace
II. taxAmount	Y	—
III. 2018tax	N	begin with numbers
IV. tax2018	Y	—
V. tax\$	N	special characters

* letters, digits, underscores I.
 * begin with letter II.
 or —
 * begins with lowercase III.
 * CANNOT contain whitespace or special characters !, #, %, etc.
 * NO C++ keywords I.

3. [2.5 pts] Identify the validity of the following statement(s). If a statement is invalid, explain why not.

Statement(s)	Validity	
	Valid/ invalid	If invalid, why?
I. const double MOR_RATE = 4.27;	Valid	—
II. const double MOR_RATE = 4.27; MOR_RATE = 3.75;	invalid	two statements - 1st is valid & 2nd is NOT
III. const double MOR_RATE;	invalid	No constant value

4. [4 pts] Which of the following are valid character literals?

* Literals are the constant values of variables.

	literal	Character Literal (Y/N)	if not a character literal, why?
I.	68	N	No Quotes / int literal
II.	"c"	N	double Quotes
III.	'c'	Y	—
IV.	"\0"	N	double Quotes / reserved char
V.	'B+'	N	Special character

5. [2 pts] You are to write a program to maintain the student scores of a class. Write statements to do each of the followings:

- Define a variable with a proper type and a proper name for the number of students in this class.
- Assign 30 to the variable. *variable = 30*

..... *const int STUDENT_NUM = 30;*
cout << "the number of students in the class: " << STUDENT_NUM << endl;

[4 pts] If income and expenditure are numeric variables, and companyName is a string variable, which of the following statements are valid assignments? If a statement is not valid, explain why not.

	Statement	Validity (Y/N)	If invalid, why?
I.	<i>income = 100;</i>	Y	—
II.	<i>expenditure = "2400.99";</i>	N	numeric variable Quoted
III.	<i>income = 31,305.52;</i>	Y	—
IV.	<i>companyName = "ABC company";</i>	Y	—
V.	<i>income * 2.1 = expenditure;</i>	N	2.1 is not defined

* technically this works, I think....
 if 2.1 was a constant float variable?
 * I think I made this mistake on my grade calc.

7. [3 pts] Write ONE statement that consolidate the following statements.

```
int flightNum = 89;
int travelTime;
int departure = 10;
int distance;
```

are these const int??

*
Two
statements

```
..... int travelTime; .....
..... distance; .....
..... const int FLIGHT_NUM = 89; .....
..... DEPARTURE = 10; .....
..... }
```

*or does
this
work?

8. [2 pts] Write down a complete C++ initialization statement to store a letter grade. Name the variable grade and set the value to be A.

```
..... char grade = 'A';
```

9. [3 pts] Complete the following table by calculating and writing the value of each C++ expression

PEMDAS

Expression	Value
$x = 5 + \frac{5+8+2=15-3=12}{8} - 3$	$x = 12$
$x = \frac{7 \cdot 4 + 5}{7} + \frac{10/2=5}{4} + \frac{10}{(5-3)}$	$x = 33$
$x = 4 + 2 * \frac{(6-2)}{8} + \frac{(4+17) \% 2}{4} - 1$	*Need help with modulo

*Can
we float
mod

?
mod
 $x \bmod y = A$
 $x/y = B$
 $B \cdot y = C$
 $C - B = A$

% = modulo

10. [2 pts] Convert the following Algebraic expressions to programming statement (you may assume the variable definition has been done priori):

$$\text{distance} = p(5rs + \frac{4}{3}r^{30})$$

*need help

```
.....
.....
.....
.....
```

11. [2.5 pts] Assume the following variable definitions and predict the output.

short LuckyNum = 2;
char initials = 'R';

* Does the value matter when finding sizeof()? *

* What kind of problems would we use sizeof? *

Statement	Output
cout << sizeof(initials) << endl;	1
cout << sizeof(LuckyNum) << endl;	2
cout << sizeof(2 * LuckyNum) << endl;	4
cout << sizeof(2 * 1.0) << endl;	8
cout << sizeof(2 * initials) << endl;	4

- Size of data type or size of operand

* What is the size it's finding?

(bytes?)

Short byte = 2

char byte = 1

* is the sizeof() measuring the bytes of the type

or data value?

Will we need to memorize the byte sizes of all data types?

12. [5 pts] Assume int x = 6, a = 2, b = 3, c = 1, d = 2. Find the value of x after the statement.

Statement	Value of x
x += 4 + b;	13
x -= 3 * d;	0
x *= 11 - c;	60
x /= b;	2
x %= d;	0

Are these consts?

* Need help/modulo

I wrote the code to find answer but still confused

What would be the output of each of the following sample codes; Assume individual codes were embedded in an error free program (#13 - #22 - 2 pts each).

13.

```
cout << "Blue " << endl << "Green" << endl << "Red";
```

output:

Blue

Green

Red

14.

```
cout << "one " << "two " << "Three";
```

output:

onetwoThree

15.

** Is this how we use multiple inputs?*

```
cout << "Enter the last three temperature readings." << endl;  
cin >> temp3 >> temp2 >> temp1;  
cout << temp1 << endl << temp2 << endl << temp3 << endl;
```

Output:

Enter the last three temperature readings.
101 100 98 [Enter]

101

100

98

16.

```
string school = "Webster";  
cout << "My School is " << "school";
```

output:

My School is Webster

17.

** Const?*

```
double number = 7.5;
cout << number << endl;
```

output:

7.5

18.

** would this be const?*

```
double tmpNumber = 7.5;
int number = tmpNumber;
cout << number << endl;
```

output:

7

19.

```
double ratio;
ratio = 5 / 2;
```

**even if it's double (data type) if there are no decimals then the output has none?
*I thought it would be 2.5 but it was 2**

```
cout << ratio << endl;
```

output:

2

20.

```
double ratio;
ratio = 5.0 / 2.0;
```

```
cout << ratio << endl;
```

output:

2.5

21.

**modulo Need help*

```
int ratio;
ratio = 5 % 2;

cout << ratio << endl;
```

output:

22. [3 pts] Write the output.

** I was confused by this format.*

Statement	output
cout << 11/ static_cast <float> (2);	5.5
cout << static_cast <char> (66);	B
cout << static_cast <float> (11/2);	5

** is that because there's two characters?
1=A 2=B 3=C ?*

23. [2 pts] Write the output of the following statement in the given space.
double distance = 62.3472;

** Need Help / confused with output*

```
cout << fixed << setprecision(2) << distance << endl
      << scientific << setprecision(3) << distance << endl;
```

output:

62.35
6.235e+01

Is this const?

24. [3 pts] Write the output of each statement.

** Need Help - scientific - what does e stand for*

Statement	Output
cout << pow(2.0, 3.0);	8
cout << abs(-6);	6
cout << sqrt(4.0);	-2

** Need help*

- I missed p1-13 chapter 3 because Dentist appt.

*page 21
chapter 3

25. [2 pts] Fill in the blank with a single statement that reads the user input given in one-line (as in the output: 470 Lockwood Ave) and store it in the variable *address*.

```
string address;
```

```
cout << "Enter the address";
```

```
.....  
getline(cin, address); Can we go over  
.....  
cout << endl << address;  
.....
```

Output:

↑ is that just for newline?

Enter the address

470 Lockwood Ave[Enter]

26. [2 pts] Write a statement to read a character, including white characters, from the keyboard and store it in the variable *continue*.

```
char continue;
```

```
cout << "Press any key to continue....";
```

**need Help*

```
.....  
.....  
.....
```

27. [4 pts] Write a C++ statement to get the following output.

You may NOT use any spaces or tabs to align the output. Use only stream manipulators: *setw()*, *fixed*, *setprecision()*, *left*, and *right*. Note: The borders and the serial numbers are just to make it easier to read; they would not be displayed in the output. Also assume the following statements already have been made.

```
double price = 2.75
```

```
count = 13,
```

```
total;
```

```
total = price * count;
```

**need help*

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
G	a	l	l	o	n		P	r	i	c	e			2	.	7	5
G	a	l	l	o	n		C	o	u	n	t					1	3
T	o	t	a	l									3	5	.	7	5

```
.....  
.....  
.....  
.....  
.....  
.....
```


Same?

30. [12 pts] Write a complete program that reads a full name as a three-word user input (i.e., first name, middle name, and last name) and store them in THREE different variables. Then display the first name and the last name on the screen.

